

120 Days of Quantitative Finance

Month 3: Machine Learning for Finance

Week 9: Supervised Machine Learning

Day 57: Loss Functions and Optimization

Arnav Garg

March 2026

1. Introduction to Supervised Learning

In supervised learning, we are given a dataset:

$$\{(x_i, y_i)\}_{i=1}^n$$

where:

- $x_i \in \mathbb{R}^p$ represents the feature vector (market signals, technical indicators, macro variables),
- $y_i \in \mathbb{R}$ (or $\{0, 1\}$) represents the target variable (returns, volatility, direction).

Our objective is to learn a function:

$$f(x; \theta)$$

parameterized by θ , such that predictions:

$$\hat{y}_i = f(x_i; \theta)$$

are as close as possible to the true targets y_i .

Thus, machine learning becomes a problem of parameter estimation through optimization.

2. Loss Functions

A loss function measures the discrepancy between the true value y and the predicted value $\hat{y}$:

$$\mathcal{L}(y, \hat{y})$$

The total empirical loss over the dataset is called the empirical risk:

$$J(\theta) = \frac{1}{n} \sum_{i=1}^n \mathcal{L}(y_i, f(x_i; \theta))$$

The learning problem becomes:

$$\theta^* = \arg \min_{\theta} J(\theta)$$

The choice of loss function determines what type of errors the model penalizes more heavily.

3. Mean Squared Error (MSE)

The most common regression loss is Mean Squared Error:

$$\mathcal{L}(y, \hat{y}) = (y - \hat{y})^2$$

Thus,

$$J(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i; \theta))^2$$

Interpretation:

- Penalizes large errors quadratically.
- Sensitive to outliers.
- Leads to closed-form solution in linear regression.
- Corresponds to Maximum Likelihood Estimation under Gaussian noise.

If:

$$y_i = f(x_i; \theta) + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$$

then minimizing MSE is equivalent to maximizing likelihood.

4. Mean Absolute Error (MAE)

Defined as:

$$\mathcal{L}(y, \hat{y}) = |y - \hat{y}|$$

Interpretation:

- Linear penalty on errors.
- More robust to outliers.
- Corresponds to Laplace-distributed noise.
- Optimization is less smooth (non-differentiable at zero).

5. Binary Cross-Entropy (Log Loss)

For classification problems where:

$$y \in \{0, 1\}$$

and predicted probability:

$$\hat{y} = P(y = 1|x)$$

the loss is:

$$\mathcal{L}(y, \hat{y}) = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$$

Interpretation:

- Penalizes confident wrong predictions heavily.
- Derived from Bernoulli likelihood.
- Used in logistic regression and neural networks.

6. Optimization Problem Structure

The general optimization problem in supervised learning is:

$$\min_{\theta} J(\theta)$$

If $J(\theta)$ is convex:

- There exists a unique global minimum.
- Gradient-based optimization converges reliably.

If $J(\theta)$ is non-convex (e.g., deep neural networks):

- Multiple local minima may exist.
- Optimization becomes more complex.

7. Bias–Variance Tradeoff

Prediction error can be decomposed as:

$$\mathbb{E}[(y - \hat{y})^2] = \text{Bias}^2 + \text{Variance} + \text{Irreducible Noise}$$

Interpretation in Finance:

- High Bias: Model too simple (underfitting).
- High Variance: Model too complex (overfitting).
- Noise: Market randomness that cannot be predicted.

The goal of supervised learning is to find an optimal balance between bias and variance.

8. Financial Interpretation

In quantitative finance, the loss function defines the investment objective:

- MSE: Minimize return prediction error.
- Asymmetric loss: Penalize downside more than upside.
- Custom loss: Optimize Sharpe ratio or drawdown.

Thus, machine learning is fundamentally objective-driven optimization.

Conclusion

Supervised machine learning reduces to minimizing a loss function over observed data. The choice of loss function determines the statistical assumptions and financial objectives of the model, while optimization determines the learned parameters. Understanding these foundations is essential before applying machine learning techniques to financial markets.